

## SIMMONS AAF, NC, USA

WMO: 746930

Lat: 35.133N

Lon: 78.933W

Elev: 74

StdP: 100.43

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93737

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |             |              |            |            | Coldest Month WS/MCDB |             |            |            | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|-------------|--------------|------------|------------|-----------------------|-------------|------------|------------|-----------------------|------------|--------------|
|               |             |             | 99.6%                         |            |             | 99%          |            |            | 0.4%                  |             | 1%         |            |                       |            |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB        | DP           | HR         | MCDB       | WS                    | MCDB        | WS         | MCDB       | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-6.0 | (c)<br>-3.8 | (d)<br>-16.3                  | (e)<br>0.9 | (f)<br>-1.8 | (g)<br>-13.4 | (h)<br>1.2 | (i)<br>0.3 | (j)<br>9.7            | (k)<br>10.3 | (l)<br>8.7 | (m)<br>9.8 | (n)<br>2.0            | (o)<br>340 | (p)<br>0.395 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.0                   | 35.6            | 24.1 | 34.2 | 24.0 | 33.0 | 23.7 | 26.0                | 31.6 | 25.5 | 30.9 | 25.0 | 30.2 | 3.5                  | 230  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 24.7                            | 20.0 | 27.5 | 24.1 | 19.2 | 27.1 | 23.7 | 18.7 | 27.0 | 80.5          | 31.6 | 78.1 | 31.1 | 76.2 | 30.1 | 29.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 8.2               | 7.1  | 5.8 | -10.3                      | 37.8 | 3.6                | 1.4 | -13.0                                              | 38.8 | -15.1      | 39.5 | -17.1      | 40.3 | -19.8      | 41.3 |
|                   |      |     | -11.3                      | 27.0 | 3.3                | 0.9 | -13.6                                              | 27.6 | -15.5      | 28.1 | -17.4      | 28.6 | -19.8      | 29.2 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 17.3   | 6.2   | 8.7   | 12.1  | 17.4  | 21.9  | 26.0  | 27.5  | 26.6  | 23.5  | 17.8  | 11.4  | 8.6   |
|                                                                        | DBStd       | 8.58   | 5.71  | 5.57  | 5.29  | 4.45  | 3.71  | 2.97  | 2.13  | 2.37  | 3.09  | 4.24  | 4.62  | 5.29  |
|                                                                        | HDD10.0     | 403    | 146   | 84    | 40    | 3     | 0     | 0     | 0     | 0     | 0     | 1     | 37    | 92    |
|                                                                        | HDD18.3     | 1518   | 377   | 273   | 204   | 70    | 13    | 1     | 0     | 0     | 2     | 62    | 212   | 305   |
|                                                                        | CDD10.0     | 3084   | 28    | 47    | 104   | 225   | 368   | 480   | 543   | 514   | 406   | 242   | 79    | 48    |
|                                                                        | CDD18.3     | 1158   | 1     | 3     | 10    | 42    | 123   | 231   | 284   | 256   | 158   | 44    | 4     | 3     |
|                                                                        | CDH23.3     | 11128  | 3     | 19    | 93    | 394   | 1093  | 2356  | 3014  | 2512  | 1264  | 346   | 22    | 11    |
|                                                                        | CDH26.7     | 4659   | 0     | 1     | 12    | 98    | 398   | 1089  | 1408  | 1104  | 468   | 81    | 1     | 1     |
| (14) Wind                                                              | WSAvg       | 2.6    | 2.6   | 2.9   | 3.1   | 3.1   | 2.7   | 2.5   | 2.4   | 2.1   | 2.3   | 2.3   | 2.4   | 2.4   |
| (15) Precipitation                                                     | PrecAvg     | 1189   | 97    | 89    | 102   | 76    | 85    | 111   | 146   | 138   | 114   | 76    | 73    | 78    |
|                                                                        | PrecMax     | 1671   | 199   | 168   | 225   | 183   | 254   | 405   | 370   | 272   | 374   | 235   | 162   | 196   |
|                                                                        | PrecMin     | 824    | 20    | 16    | 8     | 5     | 14    | 11    | 15    | 28    | 0     | 12    | 7     | 13    |
|                                                                        | PrecStd     | 180    | 44    | 44    | 46    | 41    | 46    | 69    | 68    | 64    | 82    | 53    | 42    | 36    |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 22.9  | 25.9  | 28.6  | 31.2  | 35.0  | 37.4  | 37.0  | 36.8  | 34.2  | 31.4  | 25.5  | 25.0  |
|                                                                        |             | MCWB   | 17.2  | 17.8  | 17.3  | 19.5  | 22.4  | 23.8  | 24.6  | 24.5  | 22.9  | 21.8  | 18.2  | 19.0  |
|                                                                        | 2%          | DB     | 20.4  | 22.9  | 26.1  | 29.2  | 32.4  | 35.2  | 35.3  | 35.0  | 32.6  | 28.8  | 23.6  | 22.3  |
|                                                                        |             | MCWB   | 15.8  | 15.9  | 16.9  | 19.0  | 21.5  | 23.8  | 24.6  | 24.4  | 23.0  | 21.0  | 17.1  | 18.3  |
|                                                                        | 5%          | DB     | 18.0  | 20.7  | 23.9  | 27.6  | 30.7  | 33.7  | 34.0  | 33.4  | 31.2  | 27.0  | 21.8  | 20.0  |
|                                                                        |             | MCWB   | 14.2  | 14.9  | 15.6  | 18.0  | 21.0  | 23.5  | 24.5  | 24.2  | 22.8  | 20.0  | 16.6  | 17.4  |
|                                                                        | 10%         | DB     | 16.0  | 18.3  | 21.4  | 25.8  | 28.9  | 32.2  | 32.7  | 32.0  | 29.7  | 25.1  | 19.7  | 17.6  |
|                                                                        |             | MCWB   | 12.3  | 13.6  | 14.4  | 17.4  | 20.5  | 23.1  | 24.2  | 23.9  | 22.3  | 19.4  | 15.7  | 15.1  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 19.4  | 19.5  | 19.9  | 22.4  | 24.6  | 25.8  | 26.9  | 26.6  | 25.6  | 24.7  | 20.8  | 21.2  |
|                                                                        |             | MCDB   | 21.2  | 23.1  | 24.3  | 26.4  | 29.9  | 32.3  | 32.9  | 32.7  | 29.6  | 27.0  | 22.3  | 22.6  |
|                                                                        | 2%          | WB     | 17.4  | 18.0  | 18.6  | 21.1  | 23.2  | 25.2  | 26.2  | 25.8  | 24.8  | 23.5  | 19.4  | 19.5  |
|                                                                        |             | MCDB   | 19.3  | 21.3  | 23.0  | 25.3  | 28.3  | 31.7  | 31.8  | 31.4  | 28.9  | 26.3  | 21.7  | 21.1  |
|                                                                        | 5%          | WB     | 15.3  | 16.5  | 17.5  | 20.0  | 22.4  | 24.6  | 25.6  | 25.4  | 24.1  | 22.2  | 18.0  | 17.5  |
|                                                                        |             | MCDB   | 17.2  | 19.5  | 21.2  | 25.0  | 27.5  | 30.8  | 31.1  | 30.5  | 28.2  | 25.1  | 20.6  | 19.1  |
|                                                                        | 10%         | WB     | 12.7  | 14.3  | 16.2  | 18.9  | 21.7  | 24.1  | 25.1  | 24.9  | 23.5  | 20.6  | 16.3  | 15.4  |
|                                                                        |             | MCDB   | 14.9  | 17.9  | 20.4  | 23.9  | 26.7  | 29.8  | 30.5  | 29.6  | 27.2  | 23.8  | 19.2  | 17.8  |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 10.8  | 11.8  | 12.4  | 12.4  | 11.3  | 10.5  | 10.0  | 9.9   | 9.9   | 11.6  | 12.0  | 10.7  |
|                                                                        | 5% DB       | MCDBR  | 13.6  | 14.1  | 14.5  | 14.0  | 12.8  | 12.3  | 11.2  | 11.4  | 11.7  | 12.7  | 13.7  | 12.3  |
|                                                                        |             | MCWBR  | 9.5   | 8.9   | 7.1   | 5.8   | 4.5   | 3.6   | 3.4   | 3.4   | 3.9   | 5.8   | 8.4   | 8.6   |
|                                                                        | 5% WB       | MCDBR  | 11.8  | 12.9  | 11.9  | 11.4  | 10.6  | 10.7  | 10.0  | 9.9   | 9.5   | 9.1   | 10.9  | 11.0  |
|                                                                        |             | MCWBR  | 10.1  | 9.4   | 7.5   | 6.0   | 4.3   | 3.8   | 3.5   | 3.4   | 3.6   | 5.0   | 8.5   | 8.9   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.321 | 0.334 | 0.369 | 0.408 | 0.464 | 0.501 | 0.548 | 0.524 | 0.431 | 0.391 | 0.341 | 0.319 |
|                                                                        | taud        |        | 2.481 | 2.437 | 2.374 | 2.263 | 2.149 | 2.088 | 1.975 | 2.030 | 2.290 | 2.367 | 2.466 | 2.522 |
|                                                                        | Ebn at Noon |        | 882   | 908   | 901   | 878   | 830   | 796   | 755   | 765   | 827   | 832   | 849   | 864   |
|                                                                        | Edh at Noon |        | 85    | 99    | 115   | 134   | 153   | 162   | 181   | 168   | 124   | 105   | 86    | 77    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 2.65  | 3.31  | 4.38  | 5.50  | 6.04  | 6.38  | 6.06  | 5.46  | 4.62  | 3.78  | 2.91  | 2.31  |
|                                                                        | RadStd      |        | 0.17  | 0.37  | 0.32  | 0.37  | 0.47  | 0.43  | 0.37  | 0.28  | 0.38  | 0.48  | 0.23  | 0.19  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (66 neighbors) |  | +0.46 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +157    | +113    |

Nomenclature: See separate page